Boundary Information Geometry for S-box Side-Channel Leakage: Walsh Covariance, Active Fisher Structure, and CPA Belief Dynamics

Authors and affiliations

1. **Stéphane Gaël R. Ekodeck** (corresponding author)
   ORCID: 0000-0002-8094-8832
   University of Yaounde I, LIRIMA, Team GRIMCAPE, Yaounde, Cameroon
   IRD, UMI, UMMISCO, IRD France Nord, Bondy 93143, France
   Sorbonne Universites, Univ. Paris 06, UMI 209, UMMISCO, Paris 75005, France
   Email: ekodeckstephane@gmail.com

2. **Serge Alain Ebele**
   ORCID: 0000-0001-7036-7068
   University of Yaounde I, LIRIMA, Team GRIMCAPE, Yaounde, Cameroon
   IRD, UMI, UMMISCO, IRD France Nord, Bondy 93143, France
   Sorbonne Universites, Univ. Paris 06, UMI 209, UMMISCO, Paris 75005, France
   Email: **[email address required before upload]**

3. **Prosper Rosaire Mama Assandje**
   ORCID: 0000-0002-1234-5678
   Department of Mathematics, University of Yaounde I, Yaounde, Cameroon
   Email: **[email address required before upload]**

4. **Rene Ndoundam**
   ORCID: 0000-0003-1105-762X
   University of Yaounde I, LIRIMA, Team GRIMCAPE, Yaounde, Cameroon
   IRD, UMI, UMMISCO, IRD France Nord, Bondy 93143, France
   Sorbonne Universites, Univ. Paris 06, UMI 209, UMMISCO, Paris 75005, France
   Email: **[email address required before upload]**

Corresponding author details

Dr. Stéphane Gaël R. Ekodeck
University of Yaounde I, LIRIMA, Team GRIMCAPE, Yaounde, Cameroon
Email: ekodeckstephane@gmail.com

Declarations

Competing interests

The authors declare that they have no competing interests.

Authors contributions

Conceptualization and methodology: S.G.R.E. and S.A.E.; formal analysis: S.G.R.E., S.A.E., P.R.M.A., and R.N.; software and reproducible computations: S.G.R.E.; validation: S.A.E., P.R.M.A., and R.N.; supervision and project administration: R.N.; writing–original draft: S.G.R.E.; writing–review and editing: S.G.R.E., S.A.E., P.R.M.A., and R.N. All authors read and approved the final manuscript.

Acknowledgements

This work was supported by UMMISCO, LIRIMA, and the University of Yaounde I, Cameroon. The authors used AI-assisted language tools to improve readability. After using these tools, the authors reviewed and edited the content and take full responsibility for the publication.

Authors' information

Stéphane Gaël R. Ekodeck is a senior lecturer and researcher in computer science at the University of Yaounde I, Cameroon, affiliated with LIRIMA, Team GRIMCAPE, and UMMISCO. His research interests include information security, steganography, privacy-preserving data processing, statistical modelling, and secure information systems. His recent work focuses on mathematical and information-theoretic models for secure multimedia communication, coverless steganography, and geometry-aware optimization for generative systems.

Serge Alain Ebele is a senior lecturer and researcher in computer science at the University of Yaounde I, Cameroon, affiliated with LIRIMA, Team GRIMCAPE, and UMMISCO. His research interests include complex systems, information security, discrete dynamical systems, cryptographic and steganographic models, and mathematical approaches to secure computing.

Prosper Rosaire Mama Assandje is an assistant lecturer and researcher in mathematics and information geometry affiliated with the University of Yaounde I, Cameroon. His research interests include information geometry, Hessian geometry, statistical manifolds, integrable gradient systems, Hamiltonian systems, and geometric methods for learning.

Rene Ndoundam is a full professor and researcher in computer science at the University of Yaounde I, Cameroon, affiliated with LIRIMA, Team GRIMCAPE, and UMMISCO. His research interests include theoretical computer science, cryptography, steganography, discrete dynamical systems, neural recurrence equations, distributed systems, and information security.

Article type

Research Article